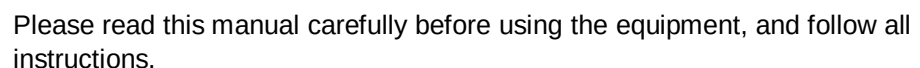


200-series loaders: A413194



Manufacturer:

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This manual uses the following warning symbols to indicate factors that must be taken into account to reduce the risk of personal injury or property damage:

	DANGER:
	This safety symbol refers to important safety information in this manual. It warns of an immediate hazard that could cause serious personal injury or property damage. Read the warning text accompanying the symbol carefully and ensure that other operators are also familiar with the warnings, since personal safety is at stake.



Note:

This notice symbol indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

In addition to the safety instructions included in this manual, you must observe all occupational safety regulations, local laws and other regulations concerning the use of the equipment. Due to continuous product development, some of the details in this manual may differ from your equipment. We reserve the right to change the contents of the manual without notification. This manual is a translation of the original Finnish version.

1. Foreword

Avant Tecno Oy would like to thank you for your purchase of this attachment for your AVANT loader. It has been designed and manufactured on the basis of years of product development and experience. By familiarising yourself with this manual and following the instructions, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Read the instructions carefully before starting to use the equipment or performing maintenance.

With these instructions, a person with basic knowledge and experience of using the loader can use the attachment safely. The manual includes important instructions for experienced AVANT operators as well. Ensure that all persons using the loader have received proper guidance and familiarised themselves with the manual of the loader and attachment and all safety instructions before using the equipment. Keep this manual at hand throughout the service life of the equipment. If the manual is lost or damaged, you can request a new one from the retailer or manufacturer. If you sell or transfer the equipment, be sure to hand over this manual to the new owner.

2. Designed purpose of use

The log splitters are attachments designed and manufactured to be suitable for use with AVANT multi purpose loaders. They are intended to be used by one person at a time for splitting firewood, whose diameter is max 400 mm. The splitting is operated by a hydraulic cylinder, which presses the wood against the splitting wedge. The height of the splitting wedge is adjustable to split the wood into halves instead of quarters. The splitter returns the cylinder after the wood has been split. Both the starting and ending points of the cylinder are adjustable according to the length of the wood. During splitting operation the operator's hands are kept on the two-hand control system of the splitter.

The attachments have been designed to require as little maintenance as possible. Regular maintenance tasks can be performed by the operator. Demanding repair and maintenance operations, such as hydraulic repairs, are to be left for professional maintenance. Familiarise yourself with the manual's instructions regarding service and maintenance. Please contact your AVANT retailer if you have additional questions about the operation or maintenance of the equipment or if you require spare parts or maintenance services.

3. Safety instructions for the log splitter

Please bear in mind that safety is the result of several factors. The loader-attachment combination is highly powerful, and improper or careless use or maintenance may cause serious personal injury or property damage. Due to this, you should carefully familiarise yourself with the manuals of both the loader and the attachment before starting operation. Do not use the attachment if you have not completely familiarised yourself with its operation and the related hazards. The purpose of these instructions is to help you to:

- use the equipment in a safe and efficient manner
 - observe and prevent any hazardous situations
 - keep the equipment intact and ensure a long service life
-
- When mounting the log splitter to the loader, ensure that the **locking pins** of the loader's quick attach plate are **in the lower position** and that they have **locked** the attachment **to the loader**.
 - The log splitter is to be used by one operator at a time. The operation of the log splitter creates a crushing hazard; do not let others near the machine when it is in use. Keep a safety distance of minimum of 2 meters.
 - When using the log splitter, keep it securely against the ground and attached to the loader. Read the instructions about the work site from section 6.
 - Always transport the log splitter as low and close to the machine as possible to keep the centre of gravity low.





- Pay attention to the contours of the terrain as well as other hazards, such as branches and trees reaching the driver's area. Pay attention to the surroundings and any other persons and machines moving in the vicinity.
- Shut down the loader engine before any maintenance or adjustments.
- Do not modify the structure of the attachment in a manner that would affect its safety.
- Do not use the log splitter to other than its intended use.
- Also read the safety instructions for the loader in the loader manual.



Familiarise yourself with the loader controls in a safe area. Pay particular attention to the safe stopping of the attachment and the machine.



Note that the loader boom moves down and may be tilted even if the engine has been stopped. Never go near the raised boom or the attachment.

Remember to wear proper protective clothing:



Never wear loose clothing, long scarves, jewellery or other items that might become entangled with the moving or rotating parts of the splitter or the wood being split. If needed, also wear a hat to prevent long hair from coming into contact with the splitter.



- The noise level in the driver's seat and the operating area may exceed 85 dB(A). Wear hearing protection while working with the loader.



- Wear safety boots while working with the log splitter.
- Wear protective gloves when handling logs or attaching the hydraulic hoses.

4. Technical specifications

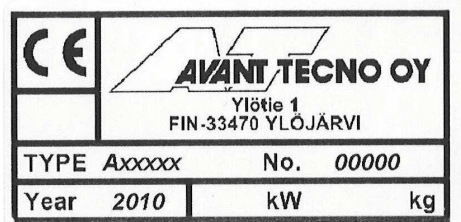
Table 1 – Log splitter technical data specifications

Product number	A413194	A13181	A413195
Log length, min/max::	200 / 650 mm		200 / 1000 mm
Log diameter, min/max:	100 / 400 mm		100 / 400 mm
Weight:	152 kg		198 kg
Minimum operating temperature:	-25°C		
Dimensions: L x W x H:	2435 x 694 x 1250 mm		3275 x 700 x 1250 mm
Supporting area:	654 x 880 mm		640 x 1040 mm
Maximum input of hydraulic energy:	60 l/min, 2,25 MPa (225 bar)		70 l/min, 2,25 MPa (226 bar)
Suitable AVANT loader:	216, 218, 220	300-700	400-700

Shown on the following page are the labels and markings, which must be visible on the equipment. Replace the warning labels, if they have become unclear or detached completely. New labels are available via your retailer or contact information provided on the cover.



The warning labels contain important safety information. Replace damaged or missing warning labels with new ones.



Identification plate of the attachment

- A46771 – Read instructions before use.
- A46772 – Do not go under a raised attachment, stay away from raised equipment.
- A46803 – Crushing hazard, stay clear of moving parts.
- A46749 – Shearing hazard, keep hands away from the splitting tools



A46771



A46772



A46803



A46749

5. Assembly of the log splitter

The log splitters are ready for use as delivered. Mount the attachment and its hydraulic hoses according to the instructions and read the following instructions for operating, adjustments and maintenance of the log splitter before starting to use the equipment.



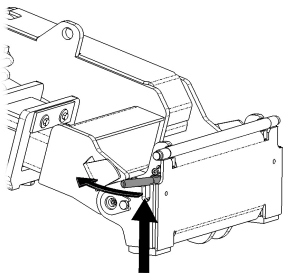
Never move or raise an attachment that has not been locked. Keep the log splitter that has not been attached at even and firm ground.

Mounting the attachment to the loader is quick and easy, but it must be done carefully. If the attachment is not locked to the loader, it may detach from the loader and cause hazardous situations. In the worst case, the attachment may slide onto the driver along a raised boom. For this reason, the attachment must under no circumstances be raised over one metre when it has not been locked. To prevent hazardous situations, always follow the attachment mounting instructions provided in the following pages. Also remember the safety instructions in Section 3.

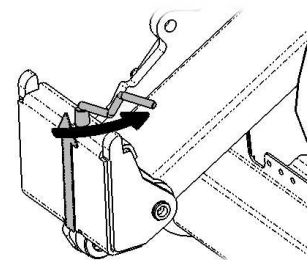
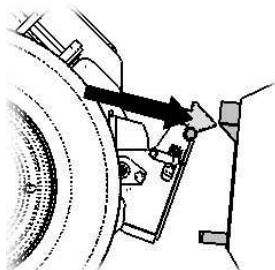
5.1. Mounting the log splitter to the loader

If your machine is equipped with the hydraulic locking and/or serial quick attach system, refer to the relevant manuals for additional mounting instructions.

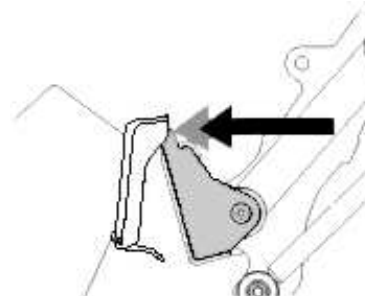
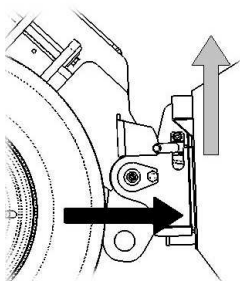
The attachment is mounted to the AVANT loader boom quick attach plate and the counterpart on the attachment. The locking is semi-automatic, but because the attachment plate of the log splitter is at relatively high level, the locking pins must be locked manually. The attachment is mounted to the loader as follows:

Step 1:

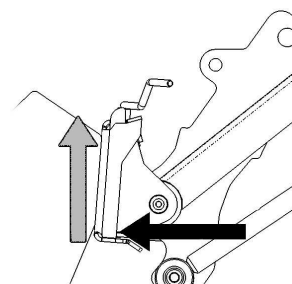
- Lift the quick attach plate locking pin / locking pins up and turn them backwards into the slot so that they are locked in the upper position.
- Ensure that the hydraulic hoses are not in the way during installation.

**Step 2:**

- Turn the quick attach plate hydraulically to an obliquely forward position.
- Drive the loader onto the attachment. If your loader is equipped with a telescopic boom, you can utilise this.

**Step 3:**

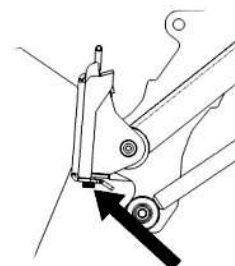
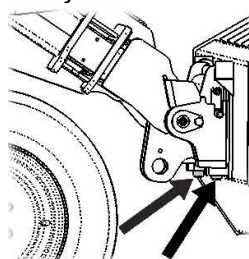
- Lift the boom slightly – pull the boom control lever backward to raise the attachment off the ground.
- Turn the boom control lever left to turn the bottom section of the quick attach plate onto the attachment.
- Lock the locking pins manually as you connect the hydraulic fittings.



Do not use the automatic return of the locking pins with the log splitter. Because the attachment plate of the log splitter is at relatively high level, tilting of an unlocked splitter increases the risk of toppling over. We recommend that you lower the log splitter onto the ground and secure the locking manually.



Before moving or raising the attachment, ensure that the locking pins are in the lower position and come through the fasteners on both sides.



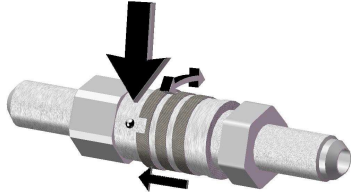
5.2. Connecting and disconnecting the hydraulic fittings

The log splitter has two hydraulic fittings, which must be connected to the corresponding fittings on the loader. When connecting or disconnecting fittings, shut down the loader engine and move the auxiliary hydraulics lever of the loader to its extreme positions a few times. This way, there will be no residual pressure in the system and the fittings are easy to connect.

To connect and disconnect the fittings, move the collar at the end of the female fitting. For correct operation of the splitter, connect the fittings with the coloured caps first, and the remaining fittings will then be fitted correctly.

In a 200-series equipment, the hose can be connected by pushing the fitting in the counterpart. When connecting and disconnecting the 200-series hydraulics quick couplings, you should pay attention to their

locking. The locking prevents the female and male fittings from being disconnected from each other by accident when the attachment moves. The fitting collar may also turn by accident into the locking position, making it impossible to connect or disconnect the fittings.



In the 200-series equipment, there is a small groove in the collar of all female fittings and a small ball below the collar. When the fittings need to be disconnected, the above-mentioned groove and ball need to be aligned. Then, the grooved collar of the female fitting is pulled downwards to disconnect the fittings from each other.



When taking the attachment off from the loader, always disconnect the hydraulic fittings before detaching the quick attach plate, to prevent hose damage and any oil spills. Reinstall the protective caps on the fittings to prevent impurities from entering the hydraulic system.



Never connect or disconnect quick couplings or other hydraulic components while the system is under pressure. High-pressure fluid injection may cause serious injuries.

Keep the fittings as clean as possible; use the protective caps for quick couplings on the attachment and loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Note that the protective caps on the loader and the attachment can be fastened to each other during operation to reduce the accumulation of dirt. Never leave the hoses hanging on the ground; place them on top of the equipment.

6. Operation

Once the log splitter has been mounted according to the instructions and the hydraulic hoses are connected, it is ready for work. Read the following instructions about work site, adjustments and use of the log splitter.

6.1. Setting up the log splitter

- Transport the log splitter carefully and note the effect of weight on the front to the stability of the loader. The splitter extension plate must be removed or folded during transportation.
- Choose a level working site, which can carry the load of the equipment and the logs. Clear the area from any obstacles that could cause a tripping hazard. The working area must not be slippery.
- Operate the log splitter in daylight or good artificial light and only in well-ventilated areas.
- Lower the splitter firmly on the ground and apply the parking brake of the loader.



Lower the log splitter firmly on the ground for use. Never go under a raised attachment or the loader boom. Note that the loader boom may be lowered or tilted even if the engine has been shut down.



Never perform any adjustments, maintenance or cleaning of the equipment if the splitter can start by accident. Shut down the loader engine first.



Never leave the equipment unattended while the engine is running.



Shut down the loader engine and release the residual pressure to prevent the splitting cylinder from moving during adjustments. If there are other persons at the work area also remove the keys from the ignition or disconnect a hydraulic fitting.

6.2. Using the splitter

Lower the attachment firmly on the ground, apply the parking brake of the loader and set moderate engine revolutions depending on the working pace. Depending on the loader model, two hydraulic pumps may be used to reduce engine noise. Note that unnecessarily high engine rotating speed will not improve the splitting force.

- Check that the log splitter is intact and that all obstacles have been removed from the working area before starting the work.
- Check the correct operation of the controls by completing a few empty cycles with the splitter. The splitting cylinder should move only when both of the control levers are actuated. After releasing the levers the cylinder should automatically return to its initial position and stop.
- Operate the attachment only from the operator's area next to the controls of the splitter.
- Ensure that the loader and the splitter are used in a safe and intended manner. Smoking near the operating area is prohibited. Do not allow children to operate the equipment.
- Check regularly to ensure there are no hydraulic leaks or faults. Refer to Section 7 for maintenance instructions.



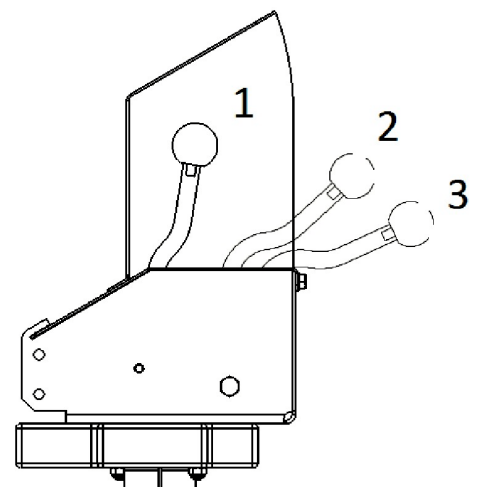
Familiarise yourself with the controls of the log splitter before using it. Pay special attention to stopping the equipment. Check before use that the controls work as intended. Never reach into the splitting area or the moving components of the equipment while it is in use; severe injuries may occur.

Turn the auxiliary hydraulics control lever on the loader to its locked position. The control device is a lever on the loader that locks to its front or side position or a pedal that locks to its lower position.

Load a log against the splitting blade, so that the squarer end of the log is placed towards the carriage plate. Never attempt to split more than one log at a time unless the carriage has been fully extended and a second log is needed to complete the separation of the first log.

Operation of the control levers, when both are operated simultaneously:

1. **Return:** The carriage will move back to its starting position as set by the setting flange. See section 6.2.1. for information about the adjustment flange.
Stopped: The carriage will stop after it reaches the initial position determined by the adjustment flange.
2. **Heavy duty:** The carriage will move towards the splitting wedge with the force of 6,5t at 180 bar.
3. **Light duty, fast forward:** The carriage will move towards the splitting wedge with the force of 4,4t at 180 bar.



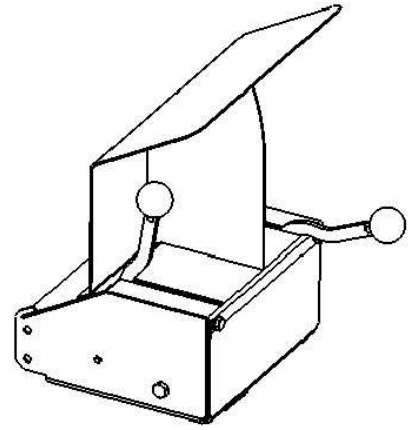
- When the levers are released, they will return to position 1 and the carriage will return to its initial position automatically.
- The levers move steplessly and the splitting speed/force can be adjusted as needed.
- Once the log starts to split the levers can be released to reduce unnecessary travel of the carriage.



To stop the splitter pull only one of the levers.



The one-lever stopping is only for temporary stopping of the cylinder and the cylinder will start its travel once the lever is released. Shut down the loader before any adjustments or removing of blockages.



In case the log gets stuck or other disruptions in use, first turn the control lever to its middle position and shut down the loader engine before other procedures. Solve the disruption after the engine has been shut down.



Always keep your fingers away from any cracks that open in the log during splitting. They can quickly close and pinch your fingers.



The log splitter is intended to be used by one operator only. Do not allow others to stay near the machine.



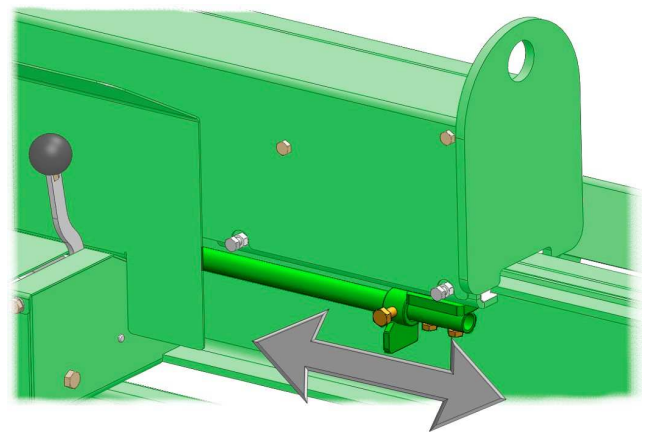
Do not use the splitter for other purposes than splitting firewood.

6.2.1. Adjusting the stroke length

When the logs are shorter than the cylinder stroke, the stroke can be adjusted so that unnecessary travel of the cylinder is minimized, thus speeding up the work.

The initial position of the stroke is adjusted by moving the adjustment flange on the push rod with tools. By moving the stopper to the left the cylinder will not return to its original starting point, making it faster to split shorter wood.

To move the flange, the cylinder should be extended slowly, and the loader engine must then be stopped.



Stopper screws for adjusting the stroke length

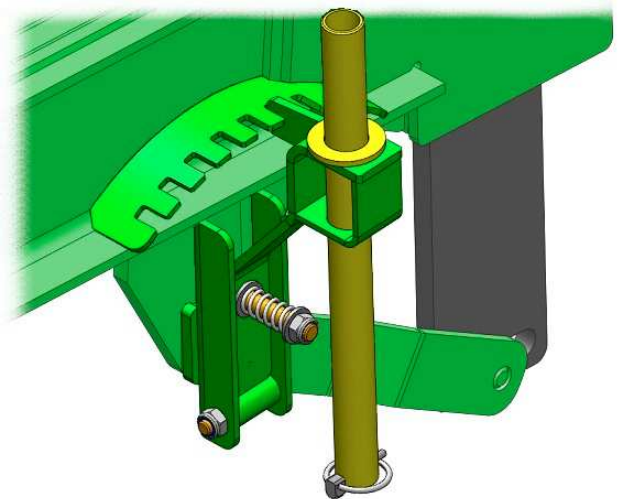


Make sure that the cylinder will not move during adjustment by shutting down the loader engine.

6.2.2. Adjusting the blade height

The height of the splitting wedge can be adjusted to match the log diameter. It can also be set to split the log to two pieces. It may be easier to split stiff wood or wood with many branches by splitting it only into two halves, so that the wood will not stick on the blade.

The adjusting bar can be extended by pulling it for easier adjustment of the height. Lock the adjustment to a suitable position in tooth rim.



Adjusting the splitting wedge

7. Maintenance and servicing

The log splitter does not require specific scheduled maintenance. Keep the equipment clean and monitor the condition of the hydraulic components. The slide pads on the carriage may be lubricated lightly and they can be adjusted to compensate normal wearing. However, the slide pads should not be set tightly against the splitter frame; the clearance should be at least 2 mm.

You may use a pressure washer and a mild detergent to wash the equipment. Avoid spraying directly at the hydraulic parts at high pressure. In addition, mind the labels on the attachment. Do not use solvents when washing.

Clean the attachment before long storage periods, touch up the paint where needed and oil the blade and other visible metal parts to prevent rust damage. The cylinder should be completely retracted during storage. Apply a small amount of oil to the visible part of the cylinder shaft.

Check the condition of the hydraulic hoses and components when the engine has been turned off and the pressure has been relieved. Repair all leaks immediately after detecting them; a small leak can quickly grow into a big one. Do not use the equipment, if you have discovered a leak in the hydraulic system. Leaking hydraulic fluid may penetrate skin and cause serious injuries. The fluid is also harmful to the environment.

Check hoses visually for cracks or abrasions. Monitor the wearing of the hoses and stop the use, if the surface layer of the hose has worn off. If there are signs of leaking, to check the component, hold up a piece of cardboard in the area where a leak is suspected. Finding any of these faults means that the hydraulic hose or component must be replaced. Spare parts are available from your nearest AVANT retailer or service point.



Never tighten a hydraulic fitting under pressure, since the fitting may break and the released oil may cause serious injuries. Do not use the equipment, if you have discovered a fault in the hydraulic system.

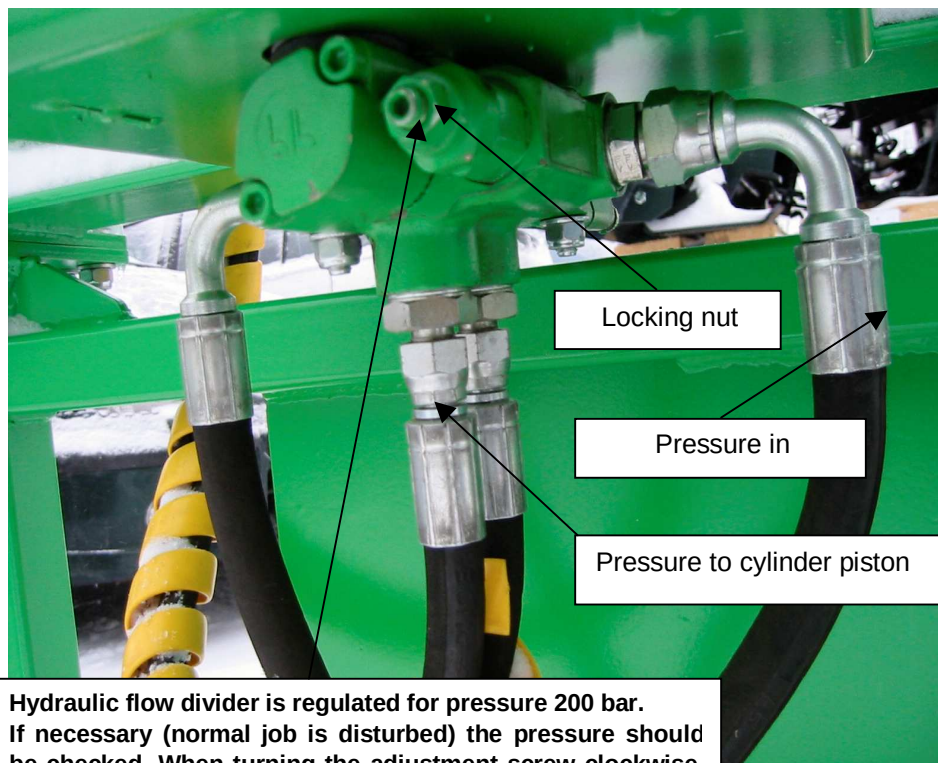


7.1. Troubleshooting

PROBLEM	POSSIBLE CAUSE	SOLUTION
No splitting action	No or too little oil	Stop immediately, check the loader hydraulic oil level and add oil
	Carriage movement incorrectly adjusted	Adjust carriage movement
	Oil too cold	Let oil circulate freely for a few minutes
	Broken or leaking hose or other hydraulic component	Stop the use, replace broken components
	Frozen	Melt and clean the equipment
Slow or weak splitting action	Oil too cold	Anna öljyn kiertää vapaasti Lisää öljyä
	Too little oil	
Log does not split	Incorrect blade position	Adjust blade position
	Large knot at splitting point	Stop splitting, turn log
	Log too large	Max diameter 40 cm
	Decreased oil pressure	Check hydraulic system (section 7.2)
Log rises up	Unsuitable log dimensions or many branches	Check carriage stroke Remove branches
Log gets jammed on blade	Incorrect stroke of carriage	Extend carriage stroke
	Blunt blade	Sharpen the blade

7.2. Hydraulic system

The hydraulic pressure of the log splitter can be checked from the flow regulator valve shown below. To check and adjust the pressure suitable equipment is needed. Contact service if the log splitter does not function properly.



Hydraulic flow divider is regulated for pressure 200 bar. If necessary (normal job is disturbed) the pressure should be checked. When turning the adjustment screw clockwise, the pressure will increase.

8. Warranty terms

Avant Tecno Oy grants a warranty of one year (12 months) from the date of purchase for the attachment it manufactures.

The warranty covers repair costs as follows:

- Work costs are covered, if the repair is not performed at the factory.
- The factory replaces any defective components or consumables.

The factory may reimburse the price of components purchased by the customer in special cases that have been agreed in advance.

The warranty does not cover:

- Normal maintenance work or parts and consumables required for it.
- Damages caused by unusual operating conditions or ways of use, negligence, overheating, structural changes made without the consent of Avant Tecno Oy, use of non-original parts or lack of maintenance.
- Consequences of a defect, such as interruption of work or other possible additional damages.
- Travel and/or freight costs caused by the repair.

EY-vaatimustenmukaisuusvakuutus EC Declaration of Conformity

Valmistaja / Manufacturer :

AS SAMI

Osoite / Address :

Tule 20

76501 Saue, HARJUMAA, ESTONIA

www.sami.ee

Vakuutamme täten, että alla mainitut tuotteet täyttävät konedirektiivin turvallisuus- ja terveystahtimukset (direktiivi 2006/42/EY muutoksineen). Seuraavia yhdenmukaistettuja standardeja on sovellettu:
EN ISO 12100, EN ISO 4413, EN 13021, EN 609-1

We hereby declare that the products listed below are in conformity with the provisions of the Machinery Directive (directive 2006/42/EC as amended). The following harmonized standards have been applied:
EN ISO 12100, EN ISO 4413, EN 13021, EN 609-1

Kuvas:

Avant-kuormainten kanssa käytettävä puunhalkaisukone

Description:

Log splitter; attachment for Avant loaders

Mallit / Models :

Avant		400-700 series
Puunhalkaisukone	PL- 650	A413181
Log splitter	PL- 1000	A413195

Allekirjoitus / Signature :

Gert Jugala,

Johtaja/
Director

20.06 2011 Saue, Estonia





AVANT[®]